Euler’s Formula (Complex Analysis)

Euler’s Formula is:

e^{iθ} = cos θ + i sin θ

1. What It Means

- e^{iθ} is a complex exponential.
- cosθ and sinθ are the real and imaginary parts.
- It links exponentials, trigonometry, and complex numbers.

2. Why It’s True (Derivation with Series)

Exponential, sine, and cosine can all be written as infinite series:

e^x = 1 + x/1! + x^2/2! + x^3/3! + ...

cos x = 1 - x^2/2! + x^4/4! - ...

sin x = x - x^3/3! + x^5/5! - ...

Plugging x = iθ into the exponential and simplifying gives exactly cosθ + i sinθ.

3. Geometric Meaning

On the complex plane, e^{iθ} lies on the unit circle, with angle θ from the positive real axis.

Examples:

e^{i0} = 1

e^{iπ/2} = i

e^{iπ} = -1

e^{i2π} = 1

4. Famous Special Case

e^{iπ} + 1 = 0

This is Euler’s Identity, linking e, i, π, 1, and 0.

5. Why It’s Useful

- Signal processing (Fourier transforms)
- Simplifies differential equations
- Represents rotations and oscillations